


```

1 import numpy as np
2 import pandas as pd
3 import os as os
4 import matplotlib.pyplot as plt
5 from PIL import Image

```

```
1 !pip install split-folders pillow ultralytics
```

Collecting split-folders

Downloading split_folders-0.6.1-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packa

Collecting ultralytics

Downloading ultralytics-8.4.75-py3-none-any.whl.metadata (41 kB)

41.3/41.3 kB 2.7 MB/s eta 0:00:

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dis

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12

Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dis

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/

Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.1

Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dis

Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/di

Requirement already satisfied: nvidia-ml-py>=12.0.0 in /usr/local/lib/python3

Collecting ultralytics-thop>=2.0.18 (from ultralytics)

Downloading ultralytics_thop-2.0.20-py3-none-any.whl.metadata (14 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3

Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/py

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/pyt

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.1

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.1

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-pac

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/py

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dis

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dis

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/d

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packa

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dis

Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/py

Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/li

Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/pyt

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/p

```

Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/l
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/lo
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/l
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/l
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/loca
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/loca
Requirement already satisfied: nvidia-cuspars-cu12==12.5.8.93.* in /usr/loca
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/loca
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac

```

```
1 !pip install roboflow
```

```
Collecting roboflow
```

```
  Downloading roboflow-1.3.10-py3-none-any.whl.metadata (11 kB)
```

```
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-pack
```

```
Collecting idna==3.7 (from roboflow)
```

```
  Downloading idna-3.7-py3-none-any.whl.metadata (9.9 kB)
```

```
Requirement already satisfied: cyclo in /usr/local/lib/python3.12/dist-packa
```

```
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12
```

```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-p
```

```
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dis
```

```
Collecting opencv-python-headless==4.10.0.84 (from roboflow)
```

```
  Downloading opencv_python_headless-4.10.0.84-cp37-abi3-manylinux_2_17_x86_6
```

```
Requirement already satisfied: Pillow>=7.1.2 in /usr/local/lib/python3.12/dis
```

```
Collecting pi-heif<2 (from roboflow)
```

```
  Downloading pi_heif-1.4.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_
```

```
Collecting pillow-avif-plugin<2 (from roboflow)
```

```
  Downloading pillow_avif_plugin-1.5.5-cp312-cp312-manylinux_2_28_x86_64.whl.
```

```
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/d
```

```
Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dis
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-pac
```

```
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
```

```
Requirement already satisfied: urllib3>=1.26.6 in /usr/local/lib/python3.12/d
```

```
Requirement already satisfied: tqdm>=4.41.0 in /usr/local/lib/python3.12/dist
```

```
Requirement already satisfied: PyYAML>=5.3.1 in /usr/local/lib/python3.12/dis
```

```
Requirement already satisfied: requests_toolbelt in /usr/local/lib/python3.12
```

```
Collecting filetype (from roboflow)
```

```
  Downloading filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)
```

```
Requirement already satisfied: typer<0.26,>=0.12.0 in /usr/local/lib/python3.
```

```
Requirement already satisfied: click>=8.0 in /usr/local/lib/python3.12/dist-p
```

```
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.1
```

```
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist
```

```
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3
```

```
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
```

```
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12
```

```
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
```

```
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
```

```
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/pyt
```

```
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python
```

```
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/pyth
```

```

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-p
Downloading roboflow-1.3.10-py3-none-any.whl (250 kB)
_____ 250.0/250.0 kB 14.0 MB/s eta 0:00
Downloading idna-3.7-py3-none-any.whl (66 kB)
_____ 66.8/66.8 kB 4.0 MB/s eta 0:00:00
Downloading opencv_python_headless-4.10.0.84-cp37-abi3-manylinux_2_17_x86_64.
_____ 49.9/49.9 MB 24.2 MB/s eta 0:00:00
Downloading pi_heif-1.4.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x8
_____ 1.5/1.5 MB 72.1 MB/s eta 0:00:00
Downloading pillow_avif_plugin-1.5.5-cp312-cp312-manylinux_2_28_x86_64.whl (5
_____ 5.5/5.5 MB 140.6 MB/s eta 0:00:00
Downloading filetype-1.2.0-py2.py3-none-any.whl (19 kB)
Installing collected packages: pillow-avif-plugin, filetype, pi-heif, opencv-
Attempting uninstall: opencv-python-headless
  Found existing installation: opencv-python-headless 4.13.0.92
  Uninstalling opencv-python-headless-4.13.0.92:
    Successfully uninstalled opencv-python-headless-4.13.0.92
Attempting uninstall: idna
  Found existing installation: idna 3.18
  Uninstalling idna-3.18:

```

```

1 from roboflow import Roboflow
2 from google.colab import userdata
3
4 roboflow_api_key = userdata.get('ROBOFLOW_API_KEY')
5 rf = Roboflow(api_key=roboflow_api_key)
6 project = rf.workspace("thaidd").project("coca-pepsi-juhuf")
7 version = project.version(5)
8 dataset = version.download("yolov11")

```

loading Roboflow workspace...

loading Roboflow project...

Downloading Dataset Version Zip in Coca-Pepsi-5 to yolov11:: 100%|██████████|

Extracting Dataset Version Zip to Coca-Pepsi-5 in yolov11:: 100%|██████████|

```

1 # Specify the path to the file
2 file_path = '/content/Coca-Pepsi-5/data.yaml'
3
4 # Check if the file exists
5 if os.path.exists(file_path):
6     print(f"The file at {file_path} exists.")
7 else:
8     print(f"The file at {file_path} does not exist.")

```

The file at /content/Coca-Pepsi-5/data.yaml exists.

```

1 # Get the current working directory
2 cwd = os.getcwd()
3 print(f"Current working directory: {cwd}")
4
5 # List all files and directories in the current directory

```

```
6 files = os.listdir(cwd)
7 print(f"Files and directories in '{cwd}': {files}")
```

Current working directory: /content
Files and directories in '/content': ['.config', 'Coca-Pepsi-5', 'sample_data']

```
1 dir_path = cwd + "/sample_data"
2 for files in os.listdir(dir_path):
3     os.remove(dir_path + "/" + files)
4 os.rmdir(dir_path)
```

```
1 from ultralytics import YOLO
2 model = YOLO("yolo11l.pt")
3 model.info()
```

Creating new Ultralytics Settings v0.0.6 file 
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.yaml'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir='/content/Coca-Pepsi-5''
Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11l.pt>
YOLO11l summary: 357 layers, 25,372,160 parameters, 0 gradients, 87.6 GFLOPs (357, 25372160, 0, 87.6134912)

```
1 from ultralytics import settings
2
3 # View all settings
4
5 settings.update({'datasets_dir': '/content/Coca-Pepsi-5'})
```

```
1 # os.chdir("/content")
2 # Entrenar el modelo
3 model.train(
4     data='/content/Coca-Pepsi-5/data.yaml',
5     epochs=50,
6     imgsz=640,
7     batch=16, # Ajusta según tu GPU
8     name="entrenamiento_logos_pepsi_cocacola",
9     project="proyecto_logos_team1"
10 )
```

Ultralytics 8.4.71 🚀 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14GB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_annot=True
Downloading <https://ultralytics.com/assets/Arial.ttf> to '/root/.config/Ultralytics/assets/Arial.ttf'
Overriding model.yaml nc=80 with nc=2

	from	n	params	module
0	-1	1	1856	ultralytics.nn.modules.conv.Conv
1	-1	1	73984	ultralytics.nn.modules.conv.Conv
2	-1	2	173824	ultralytics.nn.modules.block.C3k2
3	-1	1	590336	ultralytics.nn.modules.conv.Conv
4	-1	2	691712	ultralytics.nn.modules.block.C3k2

```

5          -1  1  2360320 ultralytics.nn.modules.conv.Conv
6          -1  2  2234368 ultralytics.nn.modules.block.C3k2
7          -1  1  2360320 ultralytics.nn.modules.conv.Conv
8          -1  2  2234368 ultralytics.nn.modules.block.C3k2
9          -1  1   656896 ultralytics.nn.modules.block.SPPF
10         -1  2  1455616 ultralytics.nn.modules.block.C2PSA
11         -1  1         0 torch.nn.modules.upsampling.Upsample
12        [-1, 6]  1         0 ultralytics.nn.modules.conv.Concat
13         -1  2  2496512 ultralytics.nn.modules.block.C3k2
14         -1  1         0 torch.nn.modules.upsampling.Upsample
15        [-1, 4]  1         0 ultralytics.nn.modules.conv.Concat
16         -1  2   756736 ultralytics.nn.modules.block.C3k2
17         -1  1   590336 ultralytics.nn.modules.conv.Conv
18        [-1, 13]  1         0 ultralytics.nn.modules.conv.Concat
19         -1  2  2365440 ultralytics.nn.modules.block.C3k2
20         -1  1  2360320 ultralytics.nn.modules.conv.Conv
21        [-1, 10]  1         0 ultralytics.nn.modules.conv.Concat
22         -1  2  2496512 ultralytics.nn.modules.block.C3k2
23        [16, 19, 22]  1  1412566 ultralytics.nn.modules.head.Detect
YOLO11l summary: 358 layers, 25,312,022 parameters, 25,312,006 gradients, 87.

```

Transferred 1009/1015 items from pretrained weights

Freezing layer 'model.23.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11l.pt>

AMP: checks passed 

train: Fast image access  (ping: 0.0±0.0 ms, read: 782.3±318.5 MB/s, size: 1.0 GB)

train: Scanning /content/Coca-Pepsi-5/train/labels... 1119 images, 6 backgrounds

train: New cache created: /content/Coca-Pepsi-5/train/labels.cache

WARNING  Box and segment counts should be equal, but got len(segments) = 69, len(boxes) = 69

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7))

val: Fast image access  (ping: 0.0±0.0 ms, read: 341.3±277.9 MB/s, size: 1.0 GB)

val: Scanning /content/Coca-Pepsi-5/valid/labels... 105 images, 0 backgrounds

val: New cache created: /content/Coca-Pepsi-5/valid/labels.cache

WARNING  Box and segment counts should be equal, but got len(segments) = 2, len(boxes) = 2

optimizer: 'optimizer=auto' found, ignoring 'lr=0.01' and 'momentum=0.937' and finding best values

optimizer: AdamW(lr=0.001667, momentum=0.9) with parameter groups 167 weight(decay=0.0), 10 no_weight_decay

Plotting labels to /content/runs/detect/proyecto_logos_team1/entrenamiento_logos_team1.jpg

Image sizes 640 train, 640 val

Using 2 dataloader workers

Logging results to /content/runs/detect/proyecto_logos_team1/entrenamiento_logos_team1.log

Starting training for 50 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/50	9.92G	1.556	1.907	1.965	47	640
	Class	Images	Instances	Box(P	R	mAP50
	all	105	128	0.00155	0.22	0.000378

```

1 # Listar archivos en el directorio
2 directory = "/content/runs/detect/proyecto_logos_team1/entrenamiento_logos_team1"
3 if os.path.exists(directory):
4     print("Archivos en el directorio:")
5     print(os.listdir(directory))
6 else:
7     print("El directorio no existe.")

```

```
7 print(f"El directorio {directory} no existe.")
```

Archivos en el directorio:

```
['weights', 'val_batch0_pred.jpg', 'BoxP_curve.png', 'val_batch0_labels.jpg',
```

```
1 # Crear DataFrame de métricas de entrenamiento
2 df_metrics = pd.read_csv("/content/runs/detect/proyecto_logos_team1/entr
3
4
5 # Mostrar algunas métricas
6 print(df_metrics.head(5))
7
8 print(df_metrics.index.to_list())
9 print(df_metrics.columns.to_list())
10
11 df_metrics.describe()
```

	epoch	time	train/box_loss	train/cls_loss	train/df1_loss \
0	1	72.9245	1.55643	1.90721	1.96543
1	2	129.9990	1.63649	1.66949	1.93249
2	3	182.6310	1.69431	1.74018	1.98929
3	4	235.3660	1.61577	1.72530	1.93883
4	5	288.8640	1.58485	1.70138	1.88846

	metrics/precision(B)	metrics/recall(B)	metrics/mAP50(B) \
0	0.00155	0.22027	0.00038
1	0.02662	0.07888	0.00665
2	0.10795	0.09518	0.03945
3	0.18589	0.26886	0.17415
4	0.17699	0.41111	0.13779

	metrics/mAP50-95(B)	val/box_loss	val/cls_loss	val/df1_loss	lr/pg0 \
0	0.00006	4.12140	6638.58000	47.10510	0.000548
1	0.00172	4.32018	21.30790	9.48488	0.001082
2	0.00999	NaN	NaN	NaN	0.001593
3	0.05395	2.63596	7.76956	3.44678	0.001568
4	0.03072	3.18727	6.17549	5.95050	0.001535

	lr/pg1	lr/pg2
0	0.000548	0.000548
1	0.001082	0.001082
2	0.001593	0.001593
3	0.001568	0.001568
4	0.001535	0.001535

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21]
['epoch', 'time', 'train/box_loss', 'train/cls_loss', 'train/df1_loss', 'metr
```

	epoch	time	train/ box_loss	train/ cls_loss	train/ df1_loss	metrics/ precision(B)	me
count	50.00000	50.000000	50.000000	50.000000	50.000000	50.000000	50
mean	25.50000	1481.050390	1.156080	1.031410	1.552525	0.736550	0
std	11.57700	850.440710	0.040450	0.001005	0.400000	0.000004	0

std	14.57738	852.449719	0.242452	0.361205	0.182333	0.239334	0
min	1.00000	72.924500	0.759050	0.486870	1.314400	0.001550	0
25%	13.25000	760.021500	1.014493	0.844348	1.416307	0.767130	0
50%	25.50000	1474.720000	1.138775	0.989225	1.502150	0.833195	0
75%	37.75000	2199.435000	1.314118	1.260935	1.651065	0.854877	0
max	50.00000	2931.920000	1.694310	1.907210	1.989290	0.928400	0

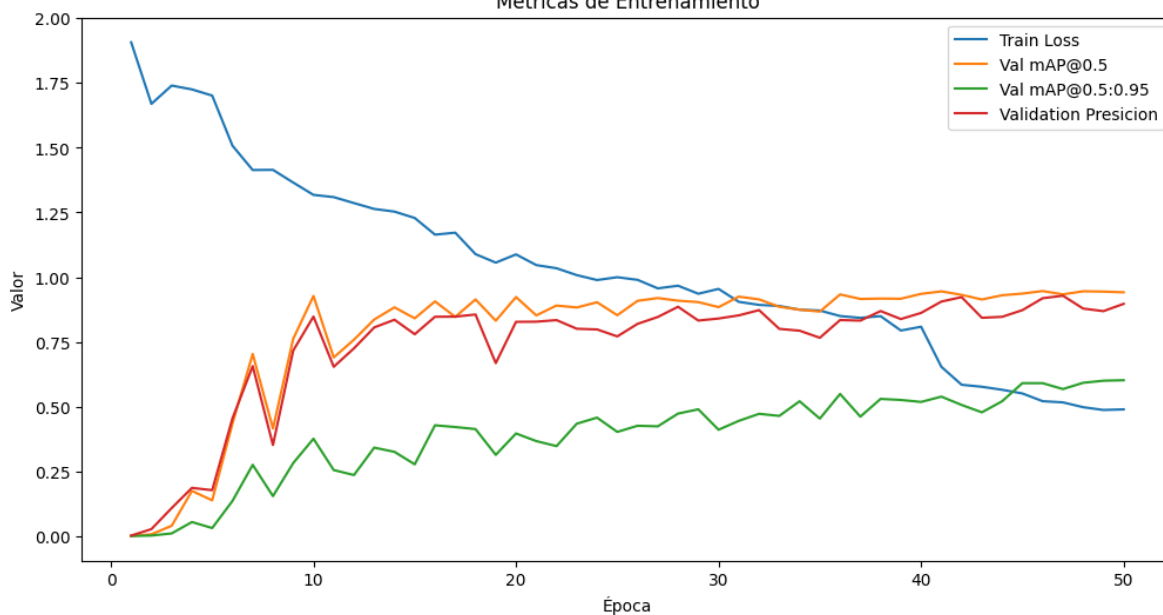
```

1 print(df_metrics.columns.to_list())
2 # Graficar las métricas
3 plt.figure(figsize=(12, 6))
4 plt.plot(df_metrics[['epoch']], df_metrics[['train/cls_loss']], label='T
5 plt.plot(df_metrics[['epoch']], df_metrics[['metrics/mAP50(B)']], label=
6 plt.plot(df_metrics[['epoch']], df_metrics[['metrics/mAP50-95(B)']], lab
7 plt.plot(df_metrics[['epoch']], df_metrics[['metrics/precision(B)']], la
8 plt.xlabel('Época')
9 plt.ylabel('Valor')
10 plt.title('Métricas de Entrenamiento')
11 plt.legend()
12 plt.show()

```

['epoch', 'time', 'train/box_loss', 'train/cls_loss', 'train/dfl_loss', 'metr

Métricas de Entrenamiento




```

1 # Evaluar el modelo
2 results = model.val()
3
4 # Mostrar los resultados
5 print(results)

```

Ultralytics 8.4.71 🚀 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14GB)
 YOLO11l summary (fused): 191 layers, 25,280,854 parameters, 0 gradients, 86.6 MB
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 978.5±309.9 MB/s, size: 24GB)
val: Scanning /content/Coca-Pepsi-5/valid/labels.cache... 105 images, 0 background
 WARNING ⚠️ Box and segment counts should be equal, but got len(segments) = 2,

Class	Images	Instances	Box(P	R	mAP50
all	105	128	0.897	0.883	0.941
CocaCola	50	65	0.882	0.862	0.945
Pepsi	59	63	0.912	0.905	0.938

 Speed: 4.7ms preprocess, 31.5ms inference, 0.0ms loss, 1.3ms postprocess per
 Results saved to /content/runs/detect/val
 ultralytics.utils.metrics.DetMetrics object with attributes:

```

ap_class_index: array([0, 1])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7e54...
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)']
curves_results: [[array([
0, 0.001001, 0.002002, 0.003003,
0.024024, 0.025025, 0.026026, 0.027027, 0.028028, 0.
0.048048, 0.049049, 0.05005, 0.051051, 0.052052, 0.
0.072072, 0.073073, 0.074074, 0.075075, 0.076076, 0.
0.096096, 0.097097, 0.098098, 0.099099, 0.1001,
0.12012, 0.12112, 0.12212, 0.12312, 0.12412, 0.
0.14414, 0.14515, 0.14615, 0.14715, 0.14815, 0.
0.16817, 0.16917, 0.17017, 0.17117, 0.17217, 0.
0.19219, 0.19319, 0.19419, 0.1952, 0.1962,
0.21622, 0.21722, 0.21822, 0.21922, 0.22022, 0.
0.24024, 0.24124, 0.24224, 0.24324, 0.24424, 0.
0.26426, 0.26527, 0.26627, 0.26727, 0.26827, 0.
0.28829, 0.28929, 0.29029, 0.29129, 0.29229, 0.
0.31231, 0.31331, 0.31431, 0.31532, 0.31632, 0.
0.33634, 0.33734, 0.33834, 0.33934, 0.34034, 0.
0.36036, 0.36136, 0.36236, 0.36336, 0.36436, 0.
0.38438, 0.38539, 0.38639, 0.38739, 0.38839, 0.
0.40841, 0.40941, 0.41041, 0.41141, 0.41241, 0.
0.43243, 0.43343, 0.43443, 0.43544, 0.43644, 0.
0.45646, 0.45746, 0.45846, 0.45946, 0.46046, 0.
0.48048, 0.48148, 0.48248, 0.48348, 0.48448, 0.
0.5045, 0.50551, 0.50651, 0.50751, 0.50851, 0.
0.52853, 0.52953, 0.53053, 0.53153, 0.53253, 0.
0.55255, 0.55355, 0.55455, 0.55556, 0.55656, 0.
0.57658, 0.57758, 0.57858, 0.57958, 0.58058, 0.

```

0.6006,	0.6016,	0.6026,	0.6036,	0.6046,	0
0.62462,	0.62563,	0.62663,	0.62763,	0.62863,	0
0.64865,	0.64965,	0.65065,	0.65165,	0.65265,	0
0.67267,	0.67367,	0.67467,	0.67568,	0.67668,	0
0.6967,	0.6977,	0.6987,	0.6997,	0.7007,	
0.72072,	0.72172,	0.72272,	0.72372,	0.72472,	0
0.74474,	0.74575,	0.74675,	0.74775,	0.74875,	0
0.76877,	0.76977,	0.77077,	0.77177,	0.77277,	0
0.79279,	0.79379,	0.79479,	0.7958,	0.7968,	
0.81682,	0.81782,	0.81882,	0.81982,	0.82082,	0
0.84084,	0.84184,	0.84284,	0.84384,	0.84484,	0
0.86486,	0.86587,	0.86687,	0.86787,	0.86887,	0
0.88889,	0.88989,	0.89089,	0.89189,	0.89289,	0
0.91291,	0.91391,	0.91491,	0.91592,	0.91692,	0
0.93694,	0.93794,	0.93894,	0.93994,	0.94094,	0
0.96096,	0.96196,	0.96296,	0.96396,	0.96496,	0

```

1 # Cargar el modelo entrenado
2 # modelo_entrenado = YOLO("/content/yolov8n.pt")
3 modelo_entrenado = YOLO("/content/runs/detect/proyecto_logos_team1/entre
4
5 # Realizar inferencia en una imagen
6 resultados = modelo_entrenado.predict(source="/content/cocacola1.jpg", s
7
8 # Acceder a los resultados
9 for resultado in resultados:
10     print(f"Predicciones para la imagen: {resultado.path}")
11
12     for caja, conf, clase in zip(resultado.bboxes.xyxy, resultado.bboxes.c
13         print(f"Bounding Box: {caja} - Confianza: {conf} - Clase: {clase

```

image 1/1 /content/cocacola1.jpg: 384x640 1 CocaCola, 97.7ms
Speed: 3.7ms preprocess, 97.7ms inference, 5.6ms postprocess per image at sha
Results saved to /content/runs/detect/predict
Predicciones para la imagen: /content/cocacola1.jpg
Bounding Box: tensor([7.9464, 0.0000, 248.0000, 148.0000], device='cuda:0

```

1 # Cargar y mostrar la imagen con las predicciones
2 img = Image.open("/content/runs/detect/predict/cocacola1.jpg")
3 plt.imshow(img)
4 plt.axis('off') # Ocultar los ejes
5 plt.show()

```





```

1 # Realizar inferencia en una imagen
2 resultados = modelo_entrenado.predict(source="/content/cocacola3.jpg", s
3
4 # Acceder a los resultados
5 for resultado in resultados:
6     print(f"Predicciones para la imagen: {resultado.path}")
7
8     for caja, conf, clase in zip(resultado.bboxes.xyxy, resultado.bboxes.c
9         print(f"Bounding Box: {caja} - Confianza: {conf} - Clase: {clase}

```

```

image 1/1 /content/cocacola3.jpg: 448x640 1 CocaCola, 95.6ms
Speed: 4.3ms preprocess, 95.6ms inference, 1.9ms postprocess per image at sha
Results saved to /content/runs/detect/predict
Predicciones para la imagen: /content/cocacola3.jpg
Bounding Box: tensor([ 0.0000,  0.6637, 140.4878, 134.2598], device='cuda:0

```

```

1 # Cargar y mostrar la imagen con las predicciones
2 img = Image.open("/content/runs/detect/predict/cocacola3.jpg")
3 plt.imshow(img)
4 plt.axis('off') # Ocultar los ejes
5 plt.show()

```



```

1 # Realizar inferencia en una imagen
2 resultados = modelo_entrenado.predict(source="/content/pepsi_cocacola2.j
3
4 # Acceder a los resultados
5 for resultado in resultados:
6     print(f"Predicciones para la imagen: {resultado.path}")
7
8     for caja, conf, clase in zip(resultado.bboxes.xyxy, resultado.bboxes.c
9         print(f"Bounding Box: {caja} - Confianza: {conf} - Clase: {clase}

```

image 1/1 /content/pepsi_cocacola2.jpg: 640x640 2 CocaColas, 1 Pepsi, 48.1ms
 Speed: 2.7ms preprocess, 48.1ms inference, 1.5ms postprocess per image at sha
 Results saved to /content/runs/detect/predict
 Predicciones para la imagen: /content/pepsi_cocacola2.jpg
 Bounding Box: tensor([85.9118, 1.0818, 142.4220, 89.2807], device='cuda:0'
 Bounding Box: tensor([5.5621, 1.1212, 67.9737, 93.4420], device='cuda:0') -
 Bounding Box: tensor([94.8988, 83.2350, 143.1685, 147.8419], device='cuda:0')

```

1 # Cargar y mostrar la imagen con las predicciones
2 img = Image.open("/content/runs/detect/predict/pepsi_cocacola2.jpg")
3 plt.imshow(img)
4 plt.axis('off') # Ocultar los ejes
5 plt.show()

```



```

1 import cv2

```

```
1 import shutil
2 from google.colab import files
3
4 # Especifica el nombre de la carpeta que deseas comprimir y descargar
5 folder_path = "/content/runs/detect/proyecto_logos_team1" # Ruta de la
6 output_filename = "juanma_pepsi_cocacola_1250_v1.zip" # Nombre del arc
7
8 # Comprimir la carpeta en un archivo ZIP
9 shutil.make_archive(output_filename.replace('.zip', ''), 'zip', folder_p
10
11 # Descargar el archivo ZIP
12 files.download(output_filename)
```